

Numerical Optimization 1

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PROBLEM I[2.1] Compute the gradient $\nabla f(x)$ and Hessian $\nabla^2 f(x)$ of the Rosenbrock function

$$f(x) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2.$$

Show that $x^* = (1, 1)^T$ is the only local minimizer of this function, and that the Hessian matrix at that point is positive definite.

PROBLEM II[2.5] Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x) = |0x|_2.$$

Show that the sequence of iterates $\{x_k\}$ defined by

$$x_k = \left(1 + \frac{1}{2^k}\right) \begin{pmatrix} \cos k \\ \sin k \end{pmatrix}$$

satisfies

$$f(x_{k+1}) < f(x_k)$$

for $k = 0, 1, 2, \dots$. Show that every point on the unit circle

$$\{x \mid |0x|_2 = 1\}$$

is a limit point for $\{x_k\}$.

Hint: Every value $\theta \in [0, 2\pi]$ is a limit point of the subsequence $\{\xi_k\}$ defined by

$$\xi_k = k \pmod{2\pi} = k - 2\pi \left\lfloor \frac{k}{2\pi} \right\rfloor,$$

where the operator $\lfloor \cdot \rfloor$ denotes rounding down to the next integer.

PROBLEM III[2.6] Prove that all isolated local minimizers are strict.

Hint: Take an isolated local minimizer x^* and a neighborhood $\mathcal{N}$. Show that for any $x \in \mathcal{N}$, $x \neq x^*$, we must have

$$f(x) > f(x^*).$$

PROBLEM IV[2.7] Suppose that

$$f(x) = x^T Q x,$$

where Q is an $n \times n$ symmetric positive semidefinite matrix. Show using the definition (1.4) that $f(x)$ is convex on the domain $\mathbb{R}^n$.

Hint: It may be convenient to prove the following equivalent inequality:

$$f(y + \alpha(x - y)) - \alpha f(x) - (1 - \alpha)f(y) \leq 0,$$

for all $\alpha \in [0, 1]$ and all $x, y \in \mathbb{R}^n$.

PROBLEM V[2.9] Consider the function

$$f(x_1, x_2) = (x_1 + x_2^2)^2.$$

At the point $x^T = (1, 0)$ we consider the search direction

$$p^T = (-1, 1).$$

Show that p is a descent direction and find all minimizers of the problem (2.10).

PROBLEM VI[2.13] (For this and the following three questions, refer to the material on “Rates of Convergence” in Section A.2 of the Appendix.)

Show that the sequence

$$x_k = \frac{1}{k}$$

is not Q-linearly convergent, though it does converge to zero. (This is called sublinear convergence.)

PROBLEM VII[2.14] Show that the sequence

$$x_k = 1 + (0.5)^{2^k}$$

is Q-quadratically convergent to 1.

PROBLEM VIII[2.15] Does the sequence

$$x_k = \frac{1}{k!}$$

converge Q-superlinearly? Q-quadratically?

PROBLEM IX[2.16] Consider the sequence $\{x_k\}$ defined by

$$x_k = \begin{cases} \left(\frac{1}{4}\right)^{2^k}, & k \text{ even,} \\ x_{k-1}/k, & k \text{ odd.} \end{cases}$$

Is this sequence Q-superlinearly convergent? Q-quadratically convergent? R-quadratically convergent?

PROBLEM X[3.1] Program the steepest descent and Newton algorithms using the backtracking line search, Algorithm 3.1. Use them to minimize the Rosenbrock function (2.22). Set the initial step

length $\alpha_0 = 1$ and print the step length used by each method at each iteration. First try the initial point

$$x_0 = (1.2, 1.2)^T$$

and then the more difficult starting point

$$x_0 = (-1.2, 1)^T.$$

PROBLEM XI[3.4] Show that the one-dimensional minimizer of a strongly convex quadratic function always satisfies the Goldstein conditions (3.11).

PROBLEM XII[3.5] Prove that

$$|0Bx|0 \geq \frac{|0x|0}{|0B^{-1}|0}$$

for any nonsingular matrix B . Use this fact to establish (3.19).

PROBLEM XIII[3.7] Prove the result (3.28) by working through the following steps.

First, use (3.26) to show that

$$|0x_k - x^*|0_Q^2 - |0x_{k+1} - x^*|0_Q^2 = 2\alpha_k \nabla f_k^T Q(x_k - x^*) - \alpha_k^2 \nabla f_k^T Q \nabla f_k,$$

where $|0 \cdot |0_Q$ is defined by (3.27).

Second, use the fact that

$$\nabla f_k = Q(x_k - x^*)$$

to obtain

$$|0x_k - x^*|0_Q^2 - |0x_{k+1} - x^*|0_Q^2 = 2 \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k} - \frac{(\nabla f_k^T \nabla f_k)^2}{\nabla f_k^T Q \nabla f_k}$$

and

$$|0x_k - x^*|0_Q^2 = \nabla f_k^T Q^{-1} \nabla f_k.$$

PROBLEM XIV[3.8] Let Q be a positive definite symmetric matrix. Prove that for any vector x , we have

$$\frac{(x^T x)^2}{(x^T Q x)(x^T Q^{-1} x)} \geq \frac{4\lambda_n \lambda_1}{(\lambda_n + \lambda_1)^2},$$

where λ_n and λ_1 are, respectively, the largest and smallest eigenvalues of Q .

(This relation, which is known as the Kantorovich inequality, can be used to deduce (3.29) from (3.28).)

PROBLEM XV[3.9] Program the BFGS algorithm using the line search algorithm described in this chapter that implements the strong Wolfe conditions. Have the code verify that

$$y_k^T s_k$$

is always positive. Use it to minimize the Rosenbrock function using the starting points given in Exercise 3.1.

PROBLEM XVI[3.12] Consider a block diagonal matrix B with 1×1 and 2×2 blocks. Show that the eigenvalues and eigenvectors of B can be obtained by computing the spectral decomposition of each diagonal block separately.